A FINITELY PRESENTED GROUP THAT IS NOT MF

ANONYMOUS

Abstract. A countable discrete group is MF here if it embeds in the unitary group of a norm matrix corona. We construct an explicit finitely presented group that is not MF, refuting the conjecture that every countable group is MF. Its reduced group C^*-algebra is a separable, stably finite C^*-algebra that is not MF, giving a concrete reduced group-algebra example rather than only an abstract existence result. The mechanism is a rigidity theorem for one-sided compressions of Kazhdan subgroups: in operator-norm matrix models, every compressor preserves the asymptotic commutant. Thus a certain commutator u converges to 1 in normalized Hilbert–Schmidt norm. A Clifford representation makes the central sign u^2 equal to -1, giving the contradiction.

More generally, every finitely presented property-(T) group with a proper injective self-embedding produces a finitely presented non-MF group. We give an explicit eight-generator, forty-one-relator instance based on $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and prove that its group rank is at most six. Its Clifford witness is a finitely generated sofic — hence hyperlinear — group that is not MF. Thus neither soficity nor hyperlinearity implies MF, and MF groups are closed under neither quotients nor extensions. Its reduced group algebra is separable, exact, faithfully tracial, stably finite, and non-MF. The finite universal Horn obstruction also defines a nonempty clopen set of non-MF marked groups.

1 Introduction

Can every countable discrete group be realized by matrices, if multiplication is only required to hold up to operator-norm errors that vanish along a sequence of finite dimensions? Shulman calls the positive assertion the *MF conjecture* [Sh], and no group was known to fail it [Sc, Sh]. Residually finite groups satisfy it through their finite quotients, as do LEF groups [CDE] and, by quasidiagonality [TWW], all amenable groups; recent work establishes it for many amalgamated free products [Sc, Sh, GKEMP]. The corresponding C^*-algebra question is older: Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable C^*-algebra was known not to be MF [BK, CDE], and their natural candidate $C^*_{\mathrm{red}}(F_2)$ was in fact proved MF by Haagerup and Thorbjørnsen [HT]. The negative solution of the Connes embedding problem subsequently supplied abstract stably finite

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non-MF algebras [JNVWY, FGH]. No countable group was known whose reduced or maximal group C^*-algebra fails to be MF [Sh].

These questions sit at the endpoint of a program of finitary approximation. In his 2018 ICM address, Thom asked for which Schatten p-norms there exist non-approximable groups [Th, Question 3.11]: the Frobenius case $p = 2$ was settled by De Chiffre–Glebsky–Lubotzky–Thom [DGLT], the cases $1 < p < \infty$ by Lubotzky–Oppenheim [LuO], and the case $p = 1$ by Bachner–Dogon–Lubotzky [BDL], who describe the three remaining metrics — the Hamming metric (soficity), the Hilbert–Schmidt norm (hyperlinearity), and the operator norm (MF) — as "the holy-grail of the subject." The Hamming case was disproved in 2026: the first nonsofic group was constructed by OpenAI [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. One-sided compression of a Kazhdan subgroup, the mechanism driving those arguments, is also the conceptual ancestor of the present one. Soficity concerns permutations in the Hamming metric, while MF concerns unitaries in operator norm, so there is no direct metric comparison between the two approximation properties. The operator-norm setting requires a different rigidity theorem, proved here by transporting finite-stage spectral corners of the adjoint actions through an equal-rank projection flip. This paper settles the operator-norm case by one explicit finitely presented group; it also proves that soficity does not imply MF and produces an explicit reduced group C^*-algebra that is stably finite but not MF. Of the three problems, only the Hilbert–Schmidt case — hyperlinearity — remains open.

Let $(d_n)_{n\in\mathbb{N}}$ be a sequence of positive integers and let

$$\mathcal{Q} \;=\; \prod_n M_{d_n}(\mathbb{C}) \Big/ \bigoplus_n M_{d_n}(\mathbb{C}) \tag{1}$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable C^*-algebra is *MF* if it embeds into some algebra of the form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman and Bachner–Dogon–Lubotzky [CDE, Sh, BDL]: a countable discrete group E is MF if, for some strictly increasing sequence of positive integers (d_n), there is an injective homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q}).$$

We shall use the equivalent coordinate model

$$\mathsf{U}_{\mathrm{cor}}((d_n)) := \prod_n \mathcal{U}(M_{d_n}(\mathbb{C})) \Big/ \{(u_n) : \|u_n - 1\| \longrightarrow 0\}. \tag{2}$$

The denominator is normal, and polar correction gives a canonical isomorphism

$$\kappa_{(d_n)} \colon \mathsf{U}_{\mathrm{cor}}((d_n)) \xrightarrow{\;\cong\;} \mathcal{U}(\mathcal{Q}), \tag{3}$$

proved in Lemma 4.2. Thus one may equivalently embed E into $\mathsf{U}_{\mathrm{cor}}((d_n))$.

Proposition 1.1 (equivalent MF formulations)**.** *For every countable group, the genuine C^*-corona definition above is equivalent to the unitary-sequence definition. The latter is equivalent to operator-norm local models with separation constant* 1*, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.* [formalized in Lean]

The proof — an explicit tensor-power amplification in one direction, a weak-MF adapter in the other, and a cumulative-block dimension normalization — is deferred, together with the precise formulation of the operator-norm local models, to Appendix A.

Hereafter "MF" means this equivalent actual-corona/local-operator-norm property. It should not be confused with stronger reduced-norm conventions. In [GKEMP], the approximating maps must additionally recover both the canonical trace and the left-regular norm on every group-ring element. The purely matricial field (PMF) convention of [MdlS, Definition 1.2] instead requires genuine finite-dimensional representations whose group-ring norms converge to the regular norms; trace convergence is included in some related conventions but not in that definition. Every assertion below uses only the MF property fixed here.

Throughout, $[g, h] = ghg^{-1}h^{-1}$ for group elements, while for matrices $[x, y] = xy - yx$; the arguments always make the reading unambiguous.

2 One-sided Kazhdan transport

The rigidity mechanism is best remembered before the construction. The same elementary principle appears in three finite worlds:

$$\begin{array}{ccc} X \subseteq Y,\ |X| = |Y| & V \subseteq W,\ \dim V = \dim W & \substack{p \le q,\ p \sim q \\ \text{in a finite } C^*\text{-algebra}} \\ \Downarrow & \Downarrow & \Downarrow \\ X = Y & V = W & p = q. \end{array}$$

Finite-dimensional representations use the middle implication directly. For matrix models, property (T) supplies at every coordinate a spectral corner — a finite-stage Kazhdan projection — and the same middle implication, in the form of an equal-rank projection flip, turns the one-sided compression of that corner into a two-sided symmetry. The right-hand implication is the classical ultraproduct shadow of this step (Remark 2.2). This yields the reusable theorem behind every result below.

We use the following terminology before collecting the remaining notation in Section 4. A *corona representation* of a group H is a homomorphism $H \to \mathcal{U}(\mathcal{Q})$, and it *detects* $g \in H$ when it does not send g to 1. An *operator-norm asymptotic representation* is a sequence of maps $U_n \colon H \to \mathcal{U}(d_n)$ such that

$$\|U_n(g)U_n(h) - U_n(gh)\| \longrightarrow 0 \qquad (g, h \in H).$$

For coordinate sequences, $a_n \sim_2 b_n$ means $\|a_n - b_n\|_2 \to 0$, where $\|x\|_2 = \operatorname{tr}_{d_n}(x^*x)^{1/2}$ is the normalized Hilbert–Schmidt norm.

Theorem 2.1 (Kazhdan transport of asymptotic commutants)**.** *Let Γ and H be groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, let $s \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$s\iota(\Gamma)s^{-1} \subseteq \iota(\Gamma).$$

Let $U_n\colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and

$$\|[x_n, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma),$$

then

$$\|[U_n(s)x_nU_n(s)^*, U_n(\iota(\gamma))]\|_2 \longrightarrow 0 \qquad (\gamma \in \Gamma).$$

[formalized in Lean]

Proof. The proof is quantitative at each single coordinate n; no ultrafilter or subsequence is needed. Property (T) provides a finite symmetric generating set $S \subseteq \Gamma$ with $1 \in S$ and a Kazhdan constant $\kappa \in (0, 1]$ for the pair (S, κ).

The adjoint model. Regard $M_{d_n}(\mathbb{C})$ as the Hilbert space $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and let each group element act by conjugation, $\operatorname{Ad} U_n(g)\,\xi = U_n(g)\,\xi\,U_n(g)^*$. Operator-norm almost multiplicativity of U_n makes $g \mapsto \operatorname{Ad} U_n(g)$ an operator-norm asymptotic representation on these Hilbert spaces; this conjugation maneuver is the finite-stage form of the $\pi \otimes \bar{\pi}$ construction.

The spectral corner. Let H_n be the Hermitian part of the average $\frac{1}{|S|}\sum_{s' \in S} \operatorname{Ad} U_n(\iota(s'))$, fix the threshold

$$\theta = \tfrac{1}{2}\Big(\Big(1 - \tfrac{\kappa^2}{4|S|}\Big) + 1\Big) < 1,$$

and let P_n be the spectral projection of H_n for the part of the spectrum above θ. Two spectral estimates tie P_n to the group.

(i) *Displacement.* For every $\gamma \in \Gamma$, $\|(\operatorname{Ad} U_n(\iota(\gamma)) - 1)\,P_n\| \to 0$: a vector whose average S-energy exceeds θ is almost fixed by each generator, because (S, κ) is a Kazhdan pair and θ exceeds $1 - \kappa^2/(4|S|)$, hence almost fixed by every word in S. The corner P_n is the finite-stage substitute for the Kazhdan projection.

(ii) *Capture.* Write $\xi_n \in L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ for the vector x_n. The commutator hypothesis says each $\operatorname{Ad} U_n(\iota(s'))$ almost fixes ξ_n, so the residual $\xi_n - H_n\xi_n$ is Hilbert–Schmidt null; a Cauchy–Schwarz energy estimate then bounds the mass of $(1 - P_n)\xi_n$ by the residual and the uniform bound on $\|x_n\|$, so ξ_n is asymptotically captured by the corner.

Leakage by the equal-rank flip. Put $T_n = \operatorname{Ad} U_n(s)$ and $Q_n = T_nP_nT_n^*$, a projection of the same rank as P_n. The compression relation enters exactly once: for $\gamma \in \Gamma$ the element s^{-1} pulls the almost-fixedness of a corner vector under $\iota(\Gamma) \supseteq s\iota(\Gamma)s^{-1}$ back to almost-fixedness under each $\iota(s')$, $s' \in S$; by the capture estimate the pulled-back vector is again almost in the corner. Thus the corner is almost contained in its conjugate, $\|(1 - Q_n)P_n\| \to 0$ — the finite-stage form of the one-sided inclusion $P \leq VPV^*$. Since $\operatorname{rank} P_n = \operatorname{rank} Q_n$, the equal-rank flip — for projections p, q of equal finite rank, $\|(1-q)p\| \le \varepsilon < 1$ implies $\|(1-p)q\| \le \varepsilon/\sqrt{1-\varepsilon^2}$ — reverses the containment:

$$\boxed{\|(1-P_n)\,Q_n\| \longrightarrow 0.}$$

This is the middle implication of the display at the head of the section, and it is the finite-dimensional substitute for finiteness of a norm ultraproduct: a one-sided compression cannot enlarge the corner.

Assembly. Fix γ and $\varepsilon > 0$, let M bound $\|x_n\|$, and put $q = \min(1, \varepsilon/(40(M^2+1)))$. For all large n: capture places ξ_n within q-mass of $P_n\xi_n$; hence $T_n\xi_n$ lies close to the range of Q_n; leakage moves it into the range of P_n; and displacement makes $\operatorname{Ad} U_n(\iota(\gamma))$ almost fix it. Bookkeeping through these three exchanges bounds the squared normalized Hilbert–Schmidt displacement of $T_n\xi_n$ by $18q^2M^2 + 16q \le \varepsilon$. Since $T_n\xi_n$ is the vector of $U_n(s)x_nU_n(s)^*$, this proves the asserted convergence — with an explicit rate in terms of κ, $|S|$, M, and the multiplicative defects of U_n on the finitely many group elements involved. □

Remark 2.2 (the ultraproduct view). The classical one-paragraph account of the same mechanism, recorded as intuition only — it is not the proof above and not the formalized route. Pass to a subsequence and a free ultrafilter ω, form the Hilbert ultraproduct K_ω of the spaces $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and the norm ultraproduct $B_\omega = \prod_\omega B(K_n)$. The adjoint actions define an honest representation π of H on K_ω, and B_ω is finite, since polar correction turns any isometry into a unitary along ω. For P the image of the Kazhdan projection [BHV] under $\pi \circ \iota$ and $V = \pi(s)$, the one-sided compression gives $P \le VPV^*$, and finiteness forces $P = VPV^*$: writing $Q = VPV^*$ and $r = V^*Q$, one has $r^*r = Q$ and $rr^* = P \le Q$, so $\sigma = r + (1-Q)$ is an isometry with $\sigma\sigma^* = P + (1-Q)$, and finiteness makes σ unitary. The finite-stage proof replaces the ultrafilter by the explicit threshold θ and replaces finiteness of B_ω by the equal-rank flip.

The negative-corner picture. The mechanism is easiest to see one level up, and this picture is the one to keep in mind. Suppose a corona representation Θ of H detects a central involution $\varepsilon = u^2$, where $u = [tct^{-1}, \iota(a)]$ is a compression defect. The nonzero projection $q = (1-\Theta(\varepsilon))/2$, central relative to $\Theta(H)$, cuts Θ to a corner representation with $\Theta_q(\varepsilon) = -q$ exactly; a spectral cut at $1/2$ of a self-adjoint lift of q produces coordinate projections whose nonzero ranks identify the corner $q\mathcal{Q}q$ with a positive-rank norm matrix corona. On that corner, c lies in the asymptotic commutant of $\iota(\Gamma)$, transport (Theorem 2.1) puts $d = tct^{-1}$ there as well, so $u \sim_2 1$ and hence $\varepsilon = u^2 \sim_2 1$ — contradicting $\varepsilon = -1$ on the corner, whose normalized Hilbert–Schmidt distance from 1 is 2. The formal development records this contradiction as a stand-alone theorem about marked operator-norm asymptotic representations; the printed proofs route instead through the finite-normal criterion of Section 9, whose coordinatewise Reynolds corner makes the corner identification sketched here rigorous without ever forming $q\mathcal{Q}q$ abstractly.

The entire non-MF argument is the following consequence of the transport theorem, applied to the two-element central subgroup $\{1, \varepsilon\}$.

Theorem 2.3 (central-sign criterion)**.** *Let H and Γ be countable groups, let Γ have property* (T)*, let $\iota\colon \Gamma \to H$ be a homomorphism, and suppose*

$$t\iota(\Gamma)t^{-1} \subseteq \iota(\Gamma), \qquad [c, \iota(\Gamma)] = 1.$$

Put $d = tct^{-1}$ and $u = [d, \iota(a)]$ for some $a \in \Gamma$. If

$$\varepsilon = u^2 \neq 1, \qquad \varepsilon^2 = 1, \qquad \varepsilon \in Z(H),$$

then every homomorphism from H to a norm matrix corona kills ε, and H is not MF. [formalized in Lean]

Proof. The data form a marked Kazhdan pattern in the sense of Definition 9.1: Γ has property (T), t compresses $\iota(\Gamma)$ into itself, and c centralizes it. Its compression-defect subgroup N_{comp} — the normal closure of the commutators $[tct^{-1}, \iota(\gamma)]$ — contains $u = [d, \iota(a)]$, hence also $\varepsilon = u^2$. Since ε is a central involution, $F = \{1, \varepsilon\}$ is a finite normal subgroup of H contained in N_{comp}. The finite-normal obstruction criterion — Theorem 9.2, proved in Section 9 from the transport theorem alone, with no result in between used in its proof — therefore kills ε under every corona representation. Since $\varepsilon \neq 1$, no corona representation of H is injective, and H is not MF. □

The negative-corner picture is the intuition; the finite-normal criterion is the engine.

3 The Kazhdan–Clifford construction

Given Γ, α, a as in the theorem below, the construction is most readable with one redundant generator ε: declare it to be a central involution and impose

$$\boxed{\varepsilon = u^2, \qquad u = [tct^{-1}, a].} \tag{4}$$

This is the conceptual presentation: rigidity forces $u \sim_2 1$, whereas the Clifford model makes $u^2 = -1$. The theorem below uses the Tietze-eliminated version. Since $d = tct^{-1}$ is an involution, $u^2 = [d, ada^{-1}]$; conversely, centrality of u^2 forces $(u^2)^2 = 1$. Thus the redundant sign costs one generator and saves the reader the only opaque word in the construction.

Theorem 3.1 (Kazhdan–Clifford construction)**.** *Let Γ be a finitely presented group with property* (T)*, let $\alpha\colon \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Adjoin elements t, c and impose*

$$t\gamma t^{-1} = \alpha(\gamma), \qquad c^2 = 1, \qquad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$d = tct^{-1}, \qquad u = [d, a], \qquad w = u^2 = [d, ada^{-1}],$$

and impose centrality of w. More explicitly, for a finite generating set S_Γ it is enough to impose the compression and lamp relations for $s \in S_\Gamma$, together with

$$[w,s]=1 \quad (s\in S_\Gamma), \qquad [w,t]=[w,c]=1.$$

Thus w commutes with a generating set of the entire resulting group. The group $E(\Gamma,\alpha,a)$ is finitely presented, the canonical map $\Gamma \to E(\Gamma,\alpha,a)$ is injective, and w is a nontrivial central involution. Every homomorphism

$$E(\Gamma,\alpha,a) \longrightarrow \mathcal{U}(\mathcal{Q})$$

to the unitary group of every norm matrix corona kills w. Consequently $E(\Gamma,\alpha,a)$ is not MF. [formalized in Lean]

Proof. Finite presentability follows from a relative finite presentation: start with the free product of Γ and the free group on t, c, and quotient by the finitely many displayed relations for a finite generating set of Γ.

The relation $w^2=1$ is redundant. Put $b=ada^{-1}$. Both d and b are involutions, $w=[d,b]=(db)^2$, and direct cancellation gives

$$dwd^{-1}=w^{-1}. \tag{5}$$

Centrality also gives $dwd^{-1}=w$, whence $w=w^{-1}$ and $w^2=1$.

For nontriviality, realize the ascending HNN extension

$$G=\langle \Gamma, t \mid t\gamma t^{-1}=\alpha(\gamma)\rangle$$

as a mapping telescope. Let Γ_∞ be the direct limit of the system $\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots$; injectivity of α passes to the limit, so the level-zero map embeds Γ in Γ_∞. Applying α levelwise induces an *automorphism* σ of Γ_∞: it is injective, and it is surjective because an element at level k equals σ of the same element at level $k+1$. Put $G=\Gamma_\infty \rtimes_\sigma \mathbb{Z}$, with stable letter t implementing σ; then $t\gamma t^{-1}=\alpha(\gamma)$ for $\gamma\in\Gamma$, and we write $X=G/\Gamma$ for the coset space of the level-zero copy. Let G permute the Clifford generators $(e_x)_{x\in X}$ and map c to e_Γ. Then

$$d \longmapsto e_{t\Gamma}, \qquad ada^{-1} \longmapsto e_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at\in\Gamma$, equivalently $a\in\alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \longmapsto (e_{t\Gamma}e_{at\Gamma})^2=-1.$$

The Clifford sign is central, so the map respects every defining relation. It follows simultaneously that $w\neq 1$ and that the canonical map of Γ is injective.

Finally, $w=u^2$ is a nontrivial central involution. The central-sign criterion therefore says that every corona representation kills w. Since $w\neq 1$, none is faithful. □

The sign argument has a useful Clifford-free strengthening. For a subgroup $L \leq H$, let $G_{\mathrm{comp}}(L)$ be generated by the elements s satisfying $sLs^{-1} \subseteq L$, and put

$$\mathfrak{D}(H, L) = \left\langle\!\!\left\langle [gzg^{-1}, \ell] : g \in G_{\mathrm{comp}}(L),\ z \in C_H(L),\ \ell \in L \right\rangle\!\!\right\rangle_H . \tag{6}$$

Theorem 9.4 in Section 9 upgrades the sign criterion to this intrinsic defect: in the presence of a property-(T) subgroup $L = \iota(\Gamma)$, every finite normal subgroup of H contained in $\mathfrak{D}(H, L)$ is killed by every corona representation of H.

The same element a drives both halves of the proof. Analytically, $a \in \Gamma$, so fixed-space transport forces the displaced lamp to commute with its image in finite-dimensional norm models. Geometrically, $a \notin \alpha(\Gamma)$, so a moves that lamp to a different HNN coset, where Clifford anticommutation produces -1. Thus a lives on the boundary of the self-compression: rigidity cannot detect that boundary, while the Clifford representation detects it by a sign.

The following theorem is the fully explicit instance, developed in Sections 5–7.

Theorem A (explicit Kazhdan–Clifford group)**.** *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the forty-one relators displayed in Definition 5.1. Put*

$$w = [tct^{-1},\ v_1(tct^{-1})v_1^{-1}].$$

The defining relations make w central; together with $c^2 = 1$, centrality implies $w^2 = 1$. Then:

(1) $w \neq 1$ in E, so w is a central involution;
(2) for every sequence (d_n) of positive integers and every group homomorphism $\Theta\colon E \to \mathsf{U}_{\mathrm{cor}}((d_n))$ into the corresponding unitary norm corona (2), one has $\Theta(w) = 1$.

In particular E is not MF, and neither $C^*_{\max}(E)$ nor $C^*_{\mathrm{red}}(E)$ is an MF C^*-algebra. [formalized in Lean]

The exponent 2 is not special. For $m \geq 2$, let E_m be the same eight-generator presentation, replacing the three relations $tv_it^{-1} = v_i^2$ by $tv_it^{-1} = v_i^m$ and leaving every other relator unchanged; thus $E_2 = E$. Write w_m for the image of the same marked word w.

Corollary 3.2 (the scaling family)**.** *For every natural number $m \geq 2$, the group E_m is finitely presented and w_m is a nontrivial central involution killed by every homomorphism from E_m to every unitary norm matrix corona. Consequently E_m is not MF, and neither $C^*_{\max}(E_m)$ nor $C^*_{\mathrm{red}}(E_m)$ is an MF C^*-algebra.* [formalized in Lean]

Proof. Replace the affine dilation $\mathrm{diag}(2, 2, 2, 1)$ by $\mathrm{diag}(m, m, m, 1)$ (both acting as in Lemma 6.2). It conjugates each integral translation v_i to v_i^m and fixes the three linear generators. Its inverse sends v_1 to translation by e_1/m, which is not integral; hence the two Clifford sites remain distinct and

w_m maps to -1. The Kazhdan base, the one-sided compression argument, and the central-sign criterion are independent of m. $\square$

For the proof, put $d = tct^{-1}$ and $u = [d, v_1]$. Lemma 7.1 gives the useful identity $w = u^2$. The proof has two opposed halves, both centered on u:

$$\boxed{\text{matrix-corona rigidity: } u \sim_2 1} \quad \text{and} \quad \boxed{\text{Clifford model: } u^2 = -1.}$$

For a corona representation that detects w, cut to the negative central corner of w. Kazhdan transport gives $u \sim_2 1$, hence $w = u^2 \sim_2 1$; the corner identity gives $w = -1$. This contradiction is the picture behind Theorem A, and it is independently formalized as a stand-alone theorem about marked asymptotic representations. Section 7 deduces the theorem from the sign criterion and the witness, and Section 9 proves the general finite-normal criterion that powers the sign criterion. [formalized in Lean]

Nontriviality of w is witnessed by a completely explicit representation of E into a semidirect product of a Clifford group by a mapping-telescope group built from the exact rational affine matrix model (Section 6); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the obstruction is elementary and needs neither property (T), nor unitarity, nor centrality of w; it kills the entire intrinsic defect (6):

Theorem B (finite-dimensional shadow)**.** *Let k be a field, V a finite-dimensional k-vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi \colon H \to \mathrm{GL}(V)$ satisfies*

$$\pi\big([\,tct^{-1},\ a(tct^{-1})a^{-1}]\big) \ = \ 1.$$

More generally, every such π sends the entire subgroup $\mathfrak{D}(H, \Gamma)$ to the identity.
[formalized in Lean]

Applied to $H = E$, Theorem B shows that *every* finite-dimensional representation of E over *every* field sends w to the identity: finite-dimensional linear representations do not separate the points of E, no finite quotient can witness $w \neq 1$, and the infinite Clifford witness of Section 6 cannot be replaced by a finite-dimensional one. In the terminology of [BDL, Definition 1.2], the central involution w, invisible in every finite quotient, makes E a group of *Deligne type*; such groups were proposed there as the natural candidate counterexamples for all three remaining approximation problems, and E realizes that proposal at the operator-norm endpoint. Property (T) enters the corona argument through a finite-stage spectral corner for the adjoint action at each coordinate, and the equal-rank projection flip turns the one-sided compression of that corner into equality.

Second, the elementary finite-dimensional calculation does not determine whether its marked word is trivial:

Theorem C (cyclic-base calibration)**.** *Define the literal marked presentation*

$$E_{\mathrm{BS}} = \left\langle \gamma_0, t, c \;\middle|\; \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\mathrm{BS}}^2 = 1, \quad [w_{\mathrm{BS}}, \gamma_0] = [w_{\mathrm{BS}}, t] = [w_{\mathrm{BS}}, c] = 1 \end{array} \right\rangle,$$

where $w_{\mathrm{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$*. Then* E_{BS} *is finitely presented and* $w_{\mathrm{BS}} \neq 1$*. There is an explicit surjective homomorphism from* E_{BS} *onto its realized Clifford quotient that maps* w_{BS} *to the nontrivial central sign. Nevertheless every finite-dimensional representation of* E_{BS} *over every field maps* w_{BS} *to the identity.* [formalized in Lean]

The quotient statement makes the comparison at the level actually used by the proof: exact finite-dimensional invisibility does not by itself force triviality of the marked word. The MF conclusion in Theorem A is the separate uniform assertion supplied by the fixed-space argument.

Generalizations and consequences Theorem A is an instance of a general finite-normal obstruction (Definition 9.1). A marked Kazhdan pattern consists of a property (T) group Λ, a homomorphism $\iota\colon \Lambda \to H$, and elements $t, c \in H$ such that $t\iota(\Lambda)t^{-1} \subseteq \iota(\Lambda)$ and $[c, \iota(\Lambda)] = 1$. Theorem 9.2 sends every finite normal subgroup contained in the resulting compression-defect subgroup to the identity under every corona representation. For E, the relevant subgroup is $\{1, w\}$. Corollary 10.5 records the resulting obstruction for C^*-subalgebras of norm matrix coronas, while Theorem C shows that finite-dimensional collapse by itself does not decide whether the marked word is trivial. The same instance yields a finite universal Horn sentence valid in every MF group and falsified by E, hence a nonempty clopen set of non-MF marked groups (Proposition 10.6 and Corollary 10.7), and the failures of quotient and extension permanence (Corollary 10.11 and Section 10).

Section 10 develops structural and C^*-algebraic consequences, including Theorem E and failures of quotient and extension permanence. Every MF algebra is stably finite [BK], so Theorem D exhibits a separable stably finite C^*-algebra that is not MF. Abstract examples follow from the failure of the Connes embedding problem [FGH]; the point here is concreteness: the algebra is the reduced group C^*-algebra of an explicit finitely presented group. Even for $C^*_{\mathrm{red}}(\mathrm{SL}_3(\mathbb{Z}))$ and $C^*_{\mathrm{red}}(\mathrm{SL}_4(\mathbb{Z}))$ the MF property is undecided; only the stronger PMF property is known to fail, for $\mathrm{SL}_4(\mathbb{Z})$ [MdlS]. Our example has the following properties:

Theorem D (the reduced group algebra of E)**.** $C^*_{\mathrm{red}}(E)$ *is a unital separable* C^**-algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.* [formalized in Lean]

The canonical homomorphism $\mathcal{B} \to E$ is injective: composing it with the homomorphism $E \to W$ of Proposition 6.3 gives $g \mapsto j(\bar{g})$, and $\mathcal{B} \to \bar{\Gamma}$ is injective by Proposition 5.2 together with Lemma 6.2. Since the infinite Kazhdan group $\mathcal{B}$ is nonamenable, so is E. Hence $C^*_{\mathrm{red}}(E)$ is not nuclear [Lan]. The nuclear form of the Blackadar–Kirchberg problem — whether every stably finite separable nuclear C^*-algebra is quasidiagonal — is therefore untouched by Theorem D. The two classical facts used in this paragraph — nonamenability of infinite Kazhdan groups and nonnuclearity of the reduced algebra of a nonamenable group — are literature results quoted for orientation only; they are outside the formal development, and no theorem below depends on them.

The witness that proves $w \neq 1$ carries the final separation:

Theorem E (sofic and hyperlinear groups that are not MF)**.** *The Clifford witness $W = \mathsf{ClLamp}(X) \rtimes V$ of Proposition 6.3 is a finitely generated sofic group that is not MF; since sofic groups are hyperlinear, W is likewise a hyperlinear group that is not MF. In particular, neither soficity nor hyperlinearity implies the MF property.* [formalized in Lean]

Operator norm versus tracial approximation The argument is specific to operator norm: the spectral-corner and leakage estimates of the transport theorem require the adjoint actions to be controlled in operator norm on the Hilbert–Schmidt spaces. Hilbert–Schmidt-small defects alone do not provide that operator control, so the proof does not give a hyperlinear obstruction.

Group-theoretic inputs The proof of Theorem A has no external group-theoretic inputs: property (T) of the literal base is proved intrinsically, from an exact rational spectral certificate through a replayed surjection and a fixed-point bridge (Proposition 5.2), and the Clifford witness and finite-dimensional obstruction are proved directly. The classical identification of the base with $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ — via the presentation of $\mathrm{SL}_3(\mathbb{Z})$ from [CRW] and property (T) of the affine group from [BHV] — is quoted as orientation only (Remark 5.3). A few later consequences quote literature results where indicated, always outside the formal claims: quasidiagonality of amenable groups [TWW] in the sharpness discussion of Section 11, the Elek–Szabó extension theorem [ES06] as context for the directly proved split-$\mathbb{Z}$ case in Theorem E, Mal'cev residual finiteness [Mal] as context for the congruence argument there, and the simple sofic embedding theorem [ES05] in the simple-envelope paragraph.

Formal verification Each gray "formalized in Lean" tag marks a closed theorem in the accompanying Lean 4 development with the same outer proposition as the printed statement. The formal track has no theorem inputs: in particular it derives property (T) of the literal twenty-relator base from an explicit rational spectral certificate rather than importing [CRW] or [BHV], and the printed proofs follow the formal dependency routes. Untagged literature consequences — the nonamenability, nonnuclearity, exactness, simple-envelope, and amenable-MF-detector paragraphs — remain deliberately outside that claim and are labelled where they occur. The public version of each tag links to the corresponding declaration on the reviewed source branch.

Relation to prior work The closest conceptual analogue is the recent work on nonsofic groups. One-sided compression of a property (T) subgroup and propagation from its centralizer appear in [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]; related centralizer rigidity for sofic embeddings is proved in [AT]. Those results concern Hamming or tracial approximations. Here the adjoint actions stay on finite-stage Hilbert–Schmidt spaces, the Kazhdan projection is replaced by explicit spectral corners, and an equal-rank projection flip prevents a one-sided compression from enlarging the corner. The complementary finite-group corner then converts that rigidity into a fixed nontrivial element sent to the identity by every corona representation. The Clifford central extension supplies the element whose survival is proved independently. The results of [OAI, KT26] provide context only: no theorem in this paper uses either one as an input.

Slofstra's constructions supply close precedents for the Clifford group, distinguished central involution, shift, and HNN doubling map [Slo18, Section 2 and Remark 3.3]. Negative-eigenspace compression appears in Slofstra–Vidick [SV, proof of Proposition 2.7], and Bachner–Dogon–Lubotzky develop operator-norm rounding, compression, and polar-correction procedures in detail [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. That local procedure enters the finite-normal corner reduction below; the transport step itself is the finite-norm-ultraproduct argument of Theorem 2.1.

The analytic passage from operator norm to the Hilbert–Schmidt conjugation representation belongs to the $\pi \otimes \bar{\pi}$ framework of Bekka and Bekka–Valette [Be, BV]. Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. These are the direct analytic precedents for the conjugation step used here.

Slofstra's quotients by elements invisible to exact or approximate finite-dimensional representations precede the MF-radical language used below [Slo17, Definitions 2.5–2.7]. Fritz's marked $\mathrm{BS}(2,3)$ example provides additional context for the cyclic comparison [Fri, SV].

Central extensions and rigidity of approximate representations also drive the Frobenius obstruction of De Chiffre–Glebsky–Lubotzky–Thom [DGLT]. In the proof of Proposition 1.6, Bachner–Dogon–Lubotzky treat the finite-central case by a character-isotypic corner, and reduce the general finite-normal case to it by passing to the finite-index centralizer [BDL]. The finite-normal criterion here combines the corresponding coordinate-corner construction with the one-sided Kazhdan compressor and does not assume that the quotient is hyperlinear. Related finite-target approximation and extension results for weak soficity appear in [GR, Gl].

Carrión–Dadarlat–Eckhardt provide the group-MF framework used here and, through Abels' group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. The cyclic comparison addresses the narrower question of what happens to the marked compression pattern when property (T) is removed.

Finally, failure of the MF property for a group C^*-algebra does not by itself prevent the group from embedding into a corona: the group property requires only a faithful canonical image in some MF quotient of $C^*_{\max}(E)$. The result here produces a specific nontrivial element in the kernel of every such representation. For broader context see [DGLT, BDL, OAI, KT26, Sh, Dad, Slo17, Slo18, SV].

Organization Sections 2–3 isolate the analytic mechanism and the general construction. Section 4 fixes notation, proves the corona lifting lemmas, and states the conventions. Section 5 constructs E and proves property (T) for its literal base. Section 6 proves $w \neq 1$. Section 7 proves the defect-square identity and deduces Theorem A from the sign criterion. Section 8 proves Theorem B. Section 9 defines marked Kazhdan patterns and their compression-defect subgroups, proves the finite-normal obstruction criterion (Theorem 9.2) and the compression radical (Theorem 9.4). Section 10 proves Theorem D, constructs a sofic non-MF witness, and establishes the permanence failures. Section 11 proves the cyclic-base calibration. Section 12 explains the operator-norm boundary of the argument. Appendix A proves the MF equivalences.

4 Notation and corona lemmas

All groups are discrete and countable. Property (T) is the textbook complex-unitary property. A C^*-homomorphism is unital only when this is stated. In particular, an MF embedding may be nonunital, as is customary for embedding a unital algebra as a C^*-subalgebra. If $e\colon A \to B$ is an injective, possibly nonunital $*$-homomorphism between unital C^*-algebras, then $u \mapsto e(u) + (1 - e(1))$ is an injective homomorphism $\mathcal{U}(A) \to \mathcal{U}(B)$; this complement correction is used below. For a unital C^*-algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r, the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \operatorname{tr}_r(x^*x)^{1/2}$; for coordinate sequences we write $a \sim_2 b$ when $\|a_n - b_n\|_2 \to 0$. We will use three elementary facts without further comment: $\|x\|_2 \le \|x\|$; $|\operatorname{tr}_r(x)| \le \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v. For a nonempty finite set Y, write $M_Y(\mathbb{C}) = \mathbb{C}^{Y\times Y}$ for the corresponding indexed matrix algebra.

Given a sequence $(d_n)_{n\in\mathbb{N}}$ of positive integers, we write

$$\mathcal{Q}((d_n)_{n\in\mathbb{N}}) \;=\; \prod_{n\in\mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \Big/ \bigoplus_{n\in\mathbb{N}}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded C^*-product by the ideal of sequences with $\|x_n\| \to 0$. We call any such algebra a *norm matrix corona*. We also write

$$\mathsf{U}_{\mathrm{cor}}((d_n)_{n\in\mathbb{N}}) = \prod_{n\in\mathbb{N}} \mathcal{U}(M_{d_n}(\mathbb{C})) \Big/ \{(u_n) : \|u_n - 1\| \longrightarrow 0\}.$$

Write $q\colon \prod_{n\in\mathbb{N}}^{\ell^\infty} M_{d_n}(\mathbb{C}) \to \mathcal{Q}$ for the quotient map of a norm matrix corona.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 4.1 (lifting)**.** *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.* [formalized in Lean]

Proof. Lift u to an arbitrary bounded sequence (x_n). Since $q((x_n))$ is unitary, the one-sided Gram defect satisfies $\|x_n^* x_n - 1\| \to 0$; no condition on $x_n x_n^*$ is needed. Once $\|x_n^* x_n - 1\| \le \frac{1}{2}$, the square Gram matrix $x_n^* x_n$ is invertible, hence so is x_n, and the polar correction $u_n = x_n(x_n^* x_n)^{-1/2}$ is unitary; continuous functional calculus gives the quantitative estimate

$$\|u_n - x_n\| \le 2\,\|x_n\|\,\|x_n^* x_n - 1\|\,.$$

The sequence (x_n) is bounded, so the right-hand side tends to 0. Setting $u_n = 1$ at the finitely many remaining indices therefore gives a unitary sequence with $\|u_n - x_n\| \to 0$. ☐

Lemma 4.2 (unitaries in the quotient)**.** *For every sequence (d_n), the map*

$$\kappa_{(d_n)}\colon \mathsf{U}_{\mathrm{cor}}((d_n)) \longrightarrow \mathcal{U}(\mathcal{Q}), \qquad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. [formalized in Lean]

Proof. The map is well defined, and its kernel consists exactly of the unitary sequences satisfying $\|u_n - 1\| \to 0$, so it is injective. It is surjective by Lemma 4.1: every unitary in $\mathcal{Q}$ has a unitary lift (u_n), whose class in (2) maps to it. ☐

A *corona representation* of a group E is a homomorphism $E \to \mathcal{U}(\mathcal{Q})$. Lemma 4.2 canonically identifies it with a homomorphism into $\mathsf{U}_{\mathrm{cor}}((d_n))$, which we call its coordinate model. The group E is *MF* if it admits an injective corona representation for a strictly increasing dimension sequence. By the dimension-amplification equivalence discussed in the introduction, allowing an arbitrary positive dimension sequence gives the same class of groups. A corona representation *detects* $g \in E$ if it does not send g to 1.

An *operator-norm asymptotic representation* of a group H is a sequence of maps $U_n\colon H \to \mathcal{U}(d_n)$, for some positive dimensions (d_n), such that

$$\|U_n(g)U_n(h) - U_n(gh)\| \longrightarrow 0 \qquad (g, h \in H).$$

No condition is imposed at the identity: taking $g = h = 1$ gives $\|U_n(1) - 1\| \to 0$ automatically, since $U_n(1)$ is unitary. Coordinate unitary lifts of a corona representation of a countable group form exactly such a sequence.

Fix a set-theoretic universe $\mathfrak{U}$. For a $\mathfrak{U}$-small group H, define $C^*_{\max,\mathfrak{U}}(H)$ to be the closed $*$-subalgebra generated by the diagonal copy of H in the bounded product of all its nontrivial unitary representations on $\mathfrak{U}$-small unital C^*-algebras. This algebra lives in the next universe. We suppress the subscript $\mathfrak{U}$ below and write u_h for the canonical unitary associated with $h \in H$.

Proposition 4.3 (universe-relative maximal group algebra)**.** *The canonical map $h \mapsto u_h$ is injective. For every $\mathfrak{U}$-small unital C^*-algebra B and every homomorphism $\rho\colon H \to \mathcal{U}(B)$, there is a unique unital $*$-homomorphism $C^*_{\max,\mathfrak{U}}(H) \to B$ mapping u_h to $\rho(h)$ for every $h \in H$.* [formalized in Lean]

For a nontrivial target, projection to its coordinate gives the extension; for a trivial target, the map is unique. Density of the generated $*$-subalgebra gives uniqueness, while the left-regular coordinate proves injectivity.

5 The literal group and its Kazhdan source

We work throughout with the groups defined by the relators below. The proof of property (T) is intrinsic to these presentations: an exact rational spectral certificate for an auxiliary Steinberg-type presentation, a replayed surjection onto the rotation presentation, and a fixed-point bridge from the rotation subgroup to the full base. No identification with a matrix group enters the proof, and every later relation involving the base's image in E is used directly from the displayed presentation. First define the *rotation group*

$$\mathcal{R} = \left\langle x, y, z \;\middle|\; \begin{array}{c} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle, \tag{7}$$

and the *literal base*

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \;\middle|\; \begin{array}{c} \text{the eight relations in (7)}, \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3,\ xv_2x^{-1} = v_1,\ xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1,\ yv_2y^{-1} = v_2^{-1}v_3,\ yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1},\ zv_2z^{-1} = v_1v_3^{-1},\ zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle. \tag{8}$$

Write $\iota\colon \mathcal{B} \to E$ for the homomorphism induced by the first six generators of the presentation below. This map is not assumed to be injective.

Definition 5.1. Use the six literal base generators and the relations displayed in (8). Define

$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \;\middle|\; \begin{array}{c} \text{all relations in (8)}, \\ tv_it^{-1} = v_i^2 \ (1 \le i \le 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle, \tag{9}$$

where w abbreviates the word $\left[\, tct^{-1},\ v_1(tct^{-1})v_1^{-1} \right]$. This is an explicit finite presentation with eight generators and forty-one relators: the twenty base relators, six stable-letter relators, c^2, six lamp relators, and eight centrality relators. Its defining relations make w central, and centrality together with $c^2 = 1$ implies $w^2 = 1$ by (5); the relations on the six base generators also give

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \qquad [c, \iota(\mathcal{B})] = 1.$$

[formalized in Lean]

Six generators suffice. The group E is generated by v_1, x, y, z, t, c; in particular its group rank is at most six. [formalized in Lean] There is also a Tietze-equivalent presentation on these six generators with thirty-two relators. In (9), eliminate v_3 and v_2 by the Tietze substitutions

$$v_3 = xv_1x^{-1}, \qquad v_2 = x^2v_1x^{-2}.$$

Delete the two relations used as definitions. After substitution, the remaining x-action relation follows from $x^3 = 1$. The two stable-letter relations for v_2, v_3 follow from the one for v_1 and $[t, x] = 1$; the two lamp relations and two centrality relations for v_2, v_3 similarly follow from those for v_1 and x. These six redundant relators may therefore be deleted. Starting from forty-one relators, the two Tietze definitions, the remaining x-action relation, and those six consequences leave thirty-two, on the generators v_1, x, y, z, t, c.

Proposition 6.3 below proves that $w \neq 1$. Figure 1 isolates the geometry used later in the analytic argument: $d = tct^{-1}$ centralizes the compressed copy, not necessarily all of $\iota(\mathcal{B})$, and the marked commutator measures precisely that failure after conjugating d by $\iota(v_1)$.

In particular the compression argument needs only the one-sided inclusion and centralizer relations shown in Figure 1; it never uses injectivity or properness of the base map. Injectivity for the constructed group is established independently by the Clifford realization.

5.1 Property (T) of the base

Proposition 5.2. *The twenty-relator group $\mathcal{B}$ has property* (T)*, in both the real-orthogonal and complex-unitary formulations.* [formalized in Lean]

Proof. The proof is intrinsic: it imports no presentation theorem and no classical property (T) statement, and it never identifies $\mathcal{B}$ with a matrix group. It has three steps.

Step 1: a certified Steinberg-type presentation. Let P_{13} be the group on the six generators $e_{12}, e_{13}, e_{21}, e_{23}, e_{31}, e_{32}$ with the thirteen relators

$$[e_{12}, e_{23}] = e_{13}, \qquad [e_{13}, e_{32}] = e_{12}, \qquad [e_{21}, e_{13}] = e_{23},$$

$$[e_{23}, e_{31}] = e_{21}, \qquad [e_{31}, e_{12}] = e_{32}, \qquad [e_{32}, e_{21}] = e_{31},$$

$$[e_{12}, e_{13}] = [e_{12}, e_{32}] = [e_{13}, e_{23}] = [e_{21}, e_{23}] = [e_{21}, e_{31}] = [e_{31}, e_{32}] = 1,$$

$$(e_{12}\, e_{21}^{-1}\, e_{12})^4 = 1,$$

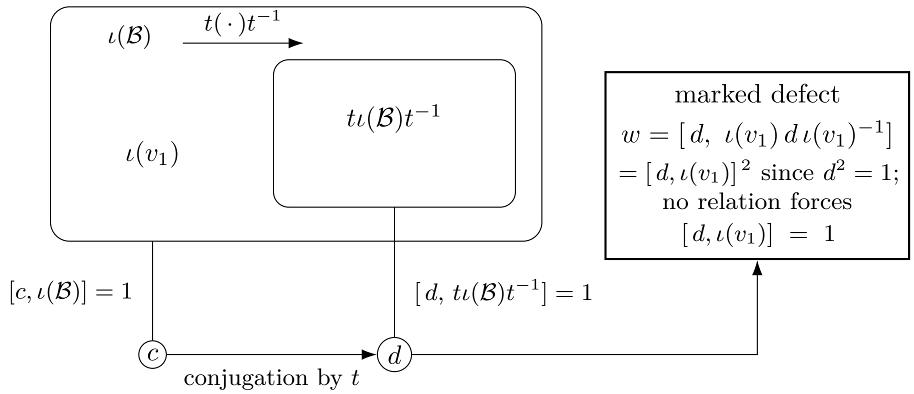

Figure 1: The marked-compression pattern. Conjugation by t carries the image of the literal base into itself. It sends c to $d = tct^{-1}$: the relations force c to commute with all of $\iota(\mathcal{B})$ and force d to commute with the compressed copy, but do not require d to commute with $\iota(v_1)$. The marked word w records that commutation defect. Only the one-sided inclusion is used by the analytic criterion; the separate Clifford witness proves that the canonical base map in this explicit group is injective.

the elementary-matrix relations in rank three. An exact sum-of-squares certificate with integer matrix data, verified by denominator-free rational arithmetic, proves the degree-one Hodge estimate for P_{13} with spectral gap $\frac{1}{500}$; testing it on coboundaries gives, in every orthogonal representation, the quadratic gap

$$\|\Delta\xi\|^2 \ \geq \ \tfrac{1}{500}\,\langle \Delta\xi, \xi\rangle$$

for the six-generator Laplacian Δ. The gap places the spectrum of Δ in $\{0\} \cup [\frac{1}{500}, \infty)$, and the kernel of Δ is exactly the space of invariant vectors; a unit vector almost fixed by the six generators has small Laplacian energy and is therefore close to an invariant vector. This yields a Kazhdan pair: P_{13} has property (T). Certificates of this kind go back to Ozawa's algebraic characterization of property (T) [Oz] and its computer-assisted implementations for $\mathrm{SL}_3(\mathbb{Z})$ [NT]; the certificate used here is exact and is replayed in full inside the formal development.

Step 2: the replayed surjection onto the rotation presentation. Map the six generators to the words

$$e_{12} \mapsto y^{-1}xzy^{-1}x^{-1}, \quad e_{13} \mapsto xzy^{-1}x^{-1}y, \quad e_{21} \mapsto y^{-1}xyxz,$$
$$e_{23} \mapsto x^{-1}y^{-1}xzy^{-1}, \quad e_{31} \mapsto xy^{-1}xzy^{-1}x, \quad e_{32} \mapsto x^{-1}zy^{-1}x^{-1}y^{-1}x^{-1}$$

in the rotation presentation $\mathcal{R}$ of (7). A word-by-word replay inside $\mathcal{R}$, using only its eight displayed relators, checks that all thirteen relators of P_{13} map to the identity, so the assignment defines a homomorphism $P_{13} \to \mathcal{R}$. The same replay exhibits explicit words in the six images whose values are z

and xy, and a further replay shows that z and xy generate $\mathcal{R}$; hence the homomorphism is surjective. Property (T) passes to quotients, so $\mathcal{R}$ has property (T).

Step 3: the fixed-point bridge to the base. The rotation generators of $\mathcal{B}$ satisfy the relators of $\mathcal{R}$, so $\mathcal{R}$ maps to the rotation subgroup of $\mathcal{B}$, and Step 2 gives a finite set S of rotation words and a $\kappa > 0$ forming a Kazhdan pair for it. Let ρ be an orthogonal representation of $\mathcal{B}$ with a unit vector x displaced by less than $\kappa/64$ by every element of the finite control set consisting of S together with the translations v_2, v_3. Three estimates produce an invariant vector near x.

(1) Projecting x to the rotation-fixed subspace gives a rotation-fixed vector p with $\|p - x\| < 1/64$, by the moving-projection estimate for the Kazhdan pair.
(2) Translations move p by a bound independent of the lattice vector. Three explicit words in the rotation generators act on the translation lattice as elementary transvections, and consequently every lattice element is a product of two rotation-conjugates of basis translations — a two-conjugate normal form proved inside the presentation. A rotation-conjugate rvr^{-1} displaces the rotation-fixed vector p exactly as much as v does, and each control translation displaces p by less than $3/64$ (transferring its bound at x across $\|p - x\|$). Hence every translation displaces p by at most $1/8$.
(3) The orbit of p under the translation subgroup is bounded, so the Hilbert-space circumcenter argument applies: the projection y of p to the translation-fixed subspace satisfies $\|y - p\| \leq 1/8$. Rotations normalize the translation subgroup and fix p, so y is also rotation-fixed; translations and rotations generate $\mathcal{B}$, so y is $\mathcal{B}$-invariant, and $\|y - x\| \leq 1/8 + 1/64 < 1$ keeps y nonzero.

Thus almost-invariant vectors produce invariant vectors and $\mathcal{B}$ has property (T) in the real-orthogonal formulation; realification and complexification identify it with the complex-unitary formulation. □

Remark 5.3 (the classical identification). Classically, the eight relations (7) present $\mathrm{SL}_3(\mathbb{Z})$ [CRW, Theorem 2] — the matrix-labelled form of the generators in Lemma 6.2 is also recorded in [CLV] — the base $\mathcal{B}$ is then the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, and property (T) of that affine group is classical [BHV, Example 1.7.4(i)]. This identification is quoted for orientation only: the proof above does not use it, and no statement in this paper depends on it.

6 The witness: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra's hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp only as an explicit algebraic witness. It uses the displayed quotient of the literal base and is independent of the semantic identification used above to prove property (T).

Construction 6.1 (Clifford lamp groups)**.** For a set X, use the real Clifford-algebra convention generated by $(c_x)_{x\in X}$ with $c_x^2 = 1$ and $c_x c_y = -c_y c_x$ for $x \neq y$, and let $\mathsf{ClLamp}(X)$ be the subgroup of units generated by -1 and the c_x. Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \qquad c_x c_y = \zeta c_y c_x \quad (x \neq y).$$

The sign ζ is nontrivial. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ. [formalized in Lean]

Lemma 6.2 (linear model)**.** *For $M \in \mathrm{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M,u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$x \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), \qquad y \mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right),$$

$$z \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right).$$

Let $\bar\Gamma \leq \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2,2,2,1)$. Direct matrix computation proves that the six matrices satisfy all twenty relators of $\mathcal{B}$ and that

$$\bar\alpha(u) = DuD^{-1}$$

is an injective endomorphism of $\bar\Gamma$ which squares the three translation generators and fixes x, y, z. Moreover the matrix assigned to v_1 does not lie in $\mathrm{range}(\bar\alpha)$*.* [formalized in Lean]

Proof. The twenty relator checks reduce to sixteen rational identities for each matrix equation. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\mathrm{GL}_4(\mathbb{Z})$. Hence $\bar\Gamma \leq \mathrm{GL}_4(\mathbb{Z})$. However $D^{-1}\bar v_1 D$ has entry $1/2$ in position $(1,4)$, so it is not in $\bar\Gamma$. This proves the last assertion. ☐

Form the mapping telescope of $\bar\alpha$, extend it by its shift τ, and call the resulting group V. Let $j\colon \bar\Gamma \hookrightarrow V$ be its level-zero map and put $X = V/j(\bar\Gamma)$, with root $o = j(\bar\Gamma)$. The final assertion of Lemma 6.2 is exactly the statement that

$$\tau o \neq j(\bar v_1)\tau o.$$

Thus the Clifford generators at these two sites anticommute.

Proposition 6.3. *$w \neq 1$ in E.* [formalized in Lean]

Proof. Form $W := \mathsf{ClLamp}(X) \rtimes V$ as in Construction 6.1. Define a map on the generators of (9) by

$$g \mapsto j(\bar{g}) \in V \quad (g \in \{v_1, v_2, v_3, x, y, z\}),$$
$$t \mapsto \tau, \qquad c \mapsto c_o.$$

The exact checks in Lemma 6.2 give the twenty base and six stable-letter relations. The level-zero subgroup fixes o, so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \qquad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two sites are distinct by Lemma 6.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W, the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \to W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E. □

Scope of the witness. The target uses the infinite group $\mathsf{ClLamp}(X)$ when X is infinite, but each group element involves only finitely many Clifford sites; the remaining data are rational matrices. No normal-form theorem for an HNN extension enters: the mapping telescope supplies the required group and the range characterization separates the two sites.

7 The defect square and the proof of Theorem A

Lemma 7.1 (the marked word is a defect square)**.** *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$\left[\, d,\ a_1\, d\, a_1^{-1} \right] \;=\; \left[\, d, a_1 \,\right]^2.$$

[formalized in Lean]

Proof. Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d\,(a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db\,db = d\,b\,d^{-1} b^{-1} = [d, b]. \qquad \square$$

Proof of Theorem A. Part (1) is Proposition 6.3. For part (2), let $\Theta\colon E \to \mathsf{U}_{\mathrm{cor}}((d_n))$ be given and pass to the corona representation $\kappa_{(d_n)} \circ \Theta\colon E \to \mathcal{U}(\mathcal{Q})$ of Lemma 4.2. Apply the central-sign criterion (Theorem 2.3) with $\Gamma = \mathcal{B}$, the canonical homomorphism ι, the elements $t, c \in E$, and $a = v_1$. Its hypotheses are exactly the displayed relators: Proposition 5.2 gives property (T) of $\mathcal{B}$; the six stable-letter relators give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$; and the six lamp relators give $[c, \iota(\mathcal{B})] = 1$. No injectivity of ι is required. In E we have $d^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 7.1 with $a_1 = \iota(v_1)$ identifies the marked word as the square of the compression defect $u = [d, \iota(v_1)]$:

$$w = \left[\, d,\ \iota(v_1)\, d\, \iota(v_1)^{-1} \right] = \left[\, d, \iota(v_1) \,\right]^2 = u^2.$$

Centrality of w is a defining relator; together with $c^2 = 1$ it forces $w^2 = 1$ by (5); and $w \neq 1$ by part (1). The criterion therefore kills w in every corona representation of E. Since $\kappa_{(d_n)}$ is injective, $\Theta(w) = 1$. This proves part (2).

For the final assertions of Theorem A: an MF embedding of E would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element w to 1. If $C^*_{\mathrm{red}}(E)$ were an MF algebra, an injective, possibly nonunital $*$-homomorphism $e\colon C^*_{\mathrm{red}}(E) \to \mathcal{Q}$ would give the injective unitary homomorphism $u \mapsto e(u) + (1 - e(1))$. Composing it with the canonical map $E \to \mathcal{U}(C^*_{\mathrm{red}}(E))$, and then with $\kappa^{-1}_{(d_n)}$, would exhibit an injective corona representation of E: the map $g \mapsto \lambda_g$ is injective. The canonical map $E \to \mathcal{U}(C^*_{\max}(E))$ is injective as well, because the left regular representation factors through $C^*_{\max}(E)$ and sends u_g to λ_g. Thus an MF embedding of $C^*_{\max}(E)$ would again produce an injective corona representation of E after the same $\kappa^{-1}_{(d_n)}$ conversion. □

The marked radical element. The proof shows more than non-injectivity of each individual Θ: the fixed element w lies in the kernel of *every* corona representation of E, so every corona representation factors through $E/\langle w\rangle$ (note $\langle w\rangle = \{1, w\}$ is central, hence normal). In the language of approximation radicals, w is a nontrivial element of the "MF radical" of E. This terminology is meant in analogy with Slofstra's earlier quotients G^{fin} and G^{fa}, obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 9.2 the same holds in every countable group with a marked Kazhdan pattern: every finite normal subgroup of the compression-defect subgroup lies in the MF radical. Section 10 develops the radical systematically; we do not know whether $E/\langle w\rangle$ is MF, i.e. whether the MF radical of E is exactly $\{1, w\}$.
No density hypothesis. No lower bound on the corner ranks r_n, or on their density in d_n, is used anywhere: the coordinate Reynolds corner in the proof of Theorem 9.2 discards zero-rank coordinates and equips each retained corner block with its own normalized trace. Thus all subsequent Hilbert–Schmidt estimates are relative to that block. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 12).

8 The finite-dimensional obstruction

We prove Theorem B. Everything happens inside the finite-dimensional algebra $\mathrm{End}_k(V)$.

Proof of Theorem B. Let $\pi\colon H \to \mathrm{GL}(V)$ be a homomorphism and let

$$C \;=\; \{\, x \in \mathrm{End}_k(V) \;:\; \pi(\gamma)\, x = x\, \pi(\gamma) \text{ for all } \gamma \in \Gamma \,\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\mathrm{End}_k(V)$. Let $u = \pi(t)$ and consider the linear automorphism $\Phi(x) = uxu^{-1}$ of $\mathrm{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and set $y = u^{-1}xu$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x, and therefore

$$\pi(\gamma)\, y = u^{-1}\, \pi(t\gamma t^{-1})\, x\, u = u^{-1}\, x\, \pi(t\gamma t^{-1})\, u = y\, \pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$C \;=\; \Phi(C) \;=\; uCu^{-1}. \tag{10}$$

Now $\pi(c) \in C$ because c centralizes Γ. By (10), $\pi(tct^{-1}) = u\, \pi(c)\, u^{-1} \in C$. Since $a \in \Gamma$, the element $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi\big(a(tct^{-1})a^{-1}\big) = \pi(tct^{-1})$, and therefore

$$\pi\big([\, tct^{-1},\ a(tct^{-1})a^{-1}]\big) = \big[\, \pi(tct^{-1}),\ \pi(tct^{-1})\big] = 1. \qquad \square$$

The same calculation proves the final, subgroup-valued assertion of Theorem B. The elements of H whose conjugation action preserves C in both directions form a subgroup. The argument above puts every one-sided compressor in that subgroup, and therefore puts all of $G_{\mathrm{comp}}(\Gamma)$ in it. Thus, for $g \in G_{\mathrm{comp}}(\Gamma)$ and $z \in C_H(\Gamma)$, the operator $\pi(gzg^{-1})$ still belongs to C and commutes with every $\pi(\ell)$, $\ell \in \Gamma$. Hence π kills all the generators in (6), and normality of its kernel kills their normal closure.

Corollary 8.1. *The group E of Definition 5.1 admits no injective homomorphism into* $\mathrm{GL}(V)$ *for any finite-dimensional vector space V over any field. More strongly, every finite-dimensional linear representation maps w to the identity, and every homomorphism to a finite group maps w to the identity.* [formalized in Lean]

Proof. Apply Theorem B with $H = E$, the image of $\mathcal{B}$, and the elements t, c, v_1; the hypotheses hold by the defining relations (9). Every finite-dimensional representation kills w, and $w \neq 1$ by Proposition 6.3. For a finite target, compose with its faithful left regular representation on the group algebra over $\mathbb{Q}$: the composite is a finite-dimensional linear representation, so it kills w, and evaluating the resulting identity matrix at the basis vector of the identity element returns the image of w in the finite group. $\square$

Why the corona argument is different. The proof of Theorem B used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. An MF model is only asymptotically multiplicative, so its coordinate fixed spaces are not literal invariant subspaces. Operator-norm control keeps the coordinate adjoint actions almost multiplicative on the Hilbert–Schmidt spaces; property (T) then supplies the spectral corner playing the role of an approximate fixed space, and the equal-rank flip turns the one-sided compression of that corner into equality — the quantitative shadow of the exact finite-dimensional argument above.

9 Kazhdan transport and finite-normal cancellation

This section proves the general finite-normal obstruction that powers the central-sign criterion and hence Theorem A. Its proof uses the transport theorem and a coordinatewise operator-norm corner, and nothing after Section 2, so the forward reference from the sign criterion is not circular. We fix the data of the criterion.

Definition 9.1 (marked Kazhdan pattern)**.** Let H be a countable group. A *marked Kazhdan pattern* in H is a tuple (Λ, ι, t, c), where Λ is a countable group with property (T), $\iota\colon \Lambda \to H$ is a group homomorphism — *not* assumed injective — and $t, c \in H$, subject to

(M1) $t\,\iota(\lambda)\,t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;
(M2) $c\,\iota(\lambda) = \iota(\lambda)\,c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The *compression-defect subgroup* of the pattern is the normal closure

$$N_{\mathrm{comp}} \;=\; \langle\!\langle\, [\, d, \iota(\lambda)\,] \;:\; \lambda \in \Lambda \,\rangle\!\rangle_H \;\trianglelefteq\; H.$$

[formalized in Lean]

Theorem 9.2 (finite-normal obstruction criterion)**.** *Let H be a countable group with a marked Kazhdan pattern (Λ, ι, t, c), and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\mathrm{comp}}$. Then every corona representation $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.* [formalized in Lean]

Proof. The argument has two independent halves. The first, which uses the Kazhdan pattern, is a kill theorem in normalized Hilbert–Schmidt norm; the second, which uses only finiteness and normality of F, converts that tracial invisibility into an exact operator-norm contradiction on a coordinate corner.

Half one: the defect dies tracially. Let $(U_{g,n})$ be coordinate unitary lifts of any operator-norm almost representation of H (in particular, of a corona representation), and put

$$K_2 = \{g \in H : \|U_{g,n} - 1\|_2 \longrightarrow 0\}.$$

Unitary invariance and asymptotic multiplicativity make $K_2 \trianglelefteq H$. Since c centralizes $\iota(\Lambda)$, its lifts belong to the asymptotic commutant of $\iota(\Lambda)$; applying Theorem 2.1 to $x_n = U_{c,n}$ puts the lifts of $d = tct^{-1}$ there as well, and therefore $[d, \iota(\lambda)] \in K_2$ for every $\lambda \in \Lambda$. Hence $N_{\mathrm{comp}} \subseteq K_2$, and so $F \subseteq K_2$: every element of the compression-defect subgroup is Hilbert–Schmidt invisible in *every* operator-norm almost representation of H.

Half two: the coordinate Reynolds corner. Suppose a corona representation Θ is nontrivial on F; choose coordinate unitary lifts $(V_{g,n})$ of Θ and form the coordinate Reynolds averages

$$p_n = \frac{1}{|F|} \sum_{f \in F} V_{f,n}.$$

Because the lifts are asymptotically multiplicative and F is finite, (p_n) is asymptotically self-adjoint and idempotent, and normality of F makes it asymptotically central for all of $\Theta(H)$. Spectral rounding at $1/2$ of the Hermitian part of p_n produces genuine coordinate projections; let q_n be their complements. Nontriviality of Θ on F forces $q_n \neq 0$ along infinitely many coordinates — otherwise the averages converge to 1 and $\Theta|_F$ is trivial. Retain exactly those coordinates, relabel them by $\mathbb{N}$, compress every lift to the corner $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$ with $r_n = \operatorname{rank} q_n > 0$, and polar-correct the compressed matrices. The result is an operator-norm almost representation $(W_{g,n})$ of H on the nonzero corner blocks; since the corner is the complement of the rounded average, the finite average vanishes there:

$$(*) \qquad \Big\| \sum_{f \in F} W_{f,n} \Big\| \longrightarrow 0 \quad \text{in operator norm.}$$

The contradiction. Half one applies to the corner almost representation $(W_{g,n})$ itself: every $f \in F$ satisfies $\|W_{f,n} - 1\|_2 \to 0$. Since F is finite, a diagonal selection of coordinates makes this uniform over F: passing to stages at which all $|F|$ distances are below $1/(n+1)$ gives

$$\sum_{f \in F} W_{f,n} \longrightarrow |F| 1 \quad \text{in normalized Hilbert–Schmidt norm,}$$

each block carrying its own normalized trace. This contradicts $(*)$, because operator-norm convergence dominates normalized Hilbert–Schmidt convergence. Hence no corona representation is nontrivial on F. □

The proof of half two used only one property of N_{comp}: every element is Hilbert–Schmidt invisible in every operator-norm almost representation of H. Any normal subgroup with that tracial-kill property therefore obstructs its finite normal subgroups in exactly the same way; the compression radical below exploits this factorization.

Theorem A(2), proved in Section 7 from the sign criterion, is also the instance $H = E$, $\Lambda = \mathcal{B}$, ι the canonical homomorphism, and $F = \{1, w\}$ of Theorem 9.2; the membership $w \in N_{\text{comp}}$ is the identity $w = [d, \iota(v_1)]^2$ of Lemma 7.1. The transport theorem is quantitative at each finite stage, so it supplies directly the ordinary sequential convergence needed above.

A uniform finite obstruction. There are a number $\delta > 0$ and a finite set $F_0 \subset E$ such that, in every matrix dimension, any unitary-valued map ϕ satisfying

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \qquad (g, h \in F_0)$$

also satisfies $\|\phi(w) - 1\| < 1$. [formalized in Lean] Thus the non-MF conclusion has a finite, dimension-free formulation that does not mention an ultrafilter. The compactness proof does not compute δ or F_0; obtaining an effective modulus directly from the rational spectral certificate remains a sharper quantitative problem.

9.1 The compression radical The transport estimate localizes at a single compressor t, but nothing in it is specific to t. Recall from (6) the subgroup $G_{\mathrm{comp}}(L)$ generated by all one-sided compressors of a subgroup $L \le H$ and the intrinsic compression–centralizer defect $\mathfrak{D}(H, L)$.

Corollary 9.3 (transport along all compressors)**.** *Let (r_n) and $V_{g,n} \in \mathcal{U}(r_n)$ be an operator-norm asymptotic representation of H, let Λ have property (T), let $\iota\colon \Lambda \to H$ be a homomorphism, and put $L = \iota(\Lambda)$. Suppose that $(x_n \in M_{r_n}(\mathbb{C}))$ is uniformly bounded in operator norm and belongs to the asymptotic commutant of L:*

$$\left\| x_n - V_{\iota(\lambda),n} x_n V_{\iota(\lambda),n}^* \right\|_2 \longrightarrow 0 \qquad (\lambda \in \Lambda).$$

Then every $g \in G_{\mathrm{comp}}(L)$ acts in both directions on that asymptotic commutant. Explicitly, both

$$V_{g,n} x_n V_{g,n}^* \quad \text{and} \quad V_{g,n}^* x_n V_{g,n}$$

satisfy the preceding convergence for every $\lambda \in \Lambda$. [formalized in Lean]

Proof. For a one-sided compressor s, Theorem 2.1 gives forward transport. Reverse transport is the same assembly run with the adjoint compressor: the equal-rank flip in that proof reverses the one-sided containment of the spectral corners, so conjugating the leakage estimate by the implementing unitary bounds the leakage for $\operatorname{Ad} U_n(s)^*$ as well, and the displacement and capture steps are unchanged. The elements acting in both directions form a subgroup of H, and hence contain the subgroup generated by all one-sided compressors, namely $G_{\mathrm{comp}}(L)$. □

Theorem 9.4 (compression radical)**.** *Let H and Γ be countable groups, let Γ have property (T), let $\iota\colon \Gamma \to H$ be a homomorphism, and put $L = \iota(\Gamma)$. If $F \trianglelefteq H$ is finite and $F \le \mathfrak{D}(H, L)$, then every homomorphism from H to every norm matrix corona kills F.* [formalized in Lean]

Proof. By the factorization recorded after Theorem 9.2, it suffices to prove the tracial half for the larger subgroup: every element of $\mathfrak{D}(H, L)$ is Hilbert–Schmidt invisible in every operator-norm almost representation of H. So let $(U_{h,n})$ be coordinate lifts of such a representation and put $K_2 = \{h : \|U_{h,n} - 1\|_2 \to 0\}$, a normal subgroup of H. If $z \in C_H(L)$, then $(U_{z,n})$ lies in the asymptotic commutant, and Corollary 9.3 puts $(U_{gzg^{-1},n})$ there for every $g \in G_{\mathrm{comp}}(L)$; hence each commutator $[gzg^{-1}, \ell]$, $\ell \in L$, belongs to K_2. Thus every displayed generator of $\mathfrak{D}(H, L)$ belongs to K_2, and normality of K_2 absorbs the normal closure, so $\mathfrak{D}(H, L) \le K_2$. The coordinate Reynolds corner and the finite-sum contradiction of Theorem 9.2 apply verbatim to any finite normal $F \le \mathfrak{D}(H, L)$. □

10 Consequences

We record the universal MF radical, the reduced-group-C^* consequence, a sofic non-MF detector, and failures of quotient and extension permanence. They all come from the same marked element.

10.1 The MF radical

Definition 10.1. For a countable group G_1, the *MF radical* is

$$\mathrm{Rad}_{\mathrm{MF}}(G_1) \;=\; \bigcap \{\ker \Theta : \;\Theta \text{ a corona representation of } G_1\},$$

a normal subgroup of G_1. This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the metric-ultraproduct approximation framework of [NST]. [formalized in Lean]

Lemma 10.2 (portability)**.** *For every homomorphism $\varphi\colon G_1 \to G_2$ of countable groups,*

$$\varphi(\mathrm{Rad}_{\mathrm{MF}}(G_1)) \;\subseteq\; \mathrm{Rad}_{\mathrm{MF}}(G_2):$$

the MF radical is functorial. In particular, if $\varphi\colon E \to G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF. [formalized in Lean]

Proof. If $x \in \mathrm{Rad}_{\mathrm{MF}}(G_1)$ and Θ is a corona representation of G_2, then $\Theta \circ \varphi$ is a corona representation of G_1, so $\Theta(\varphi(x)) = 1$. For the second statement: $w \in \mathrm{Rad}_{\mathrm{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \mathrm{Rad}_{\mathrm{MF}}(G_2)$, and an injective corona representation of G_2 would kill $\varphi(w)$, a contradiction. □

Proposition 10.3 (the universal MF quotient)**.** *Let G_1 be a countable group and $R = \mathrm{Rad}_{\mathrm{MF}}(G_1)$.*

(1) R is the kernel of a single corona representation of G_1.
(2) G_1/R is MF.
(3) Every homomorphism from G_1 to an MF group factors through G_1/R.

In particular G_1 is MF if and only if $R = \{1\}$. [formalized in Lean]

The statement is presumably folklore once the radical is defined; the proof is included for completeness, with no claim of novelty.

Proof. (3) If $\varphi\colon G_1 \to M$ with M MF and $j\colon M \to \mathsf{U}_{\mathrm{cor}}((d_n))$ is an injective corona representation, then $j \circ \varphi$ is a corona representation of G_1, so $j(\varphi(R)) = 1$ and, by injectivity of j, $\varphi(R) = 1$.

(1) and (2). Work on $Q = G_1/R$ from the outset. If $1 \neq q \in Q$ and $g \in G_1$ represents q, then $g \notin R$. Hence some corona representation of G_1 detects g. Every corona representation kills R by definition, so this representation descends to one of Q that detects q. Thus the intersection of the kernels of all corona representations of the countable group Q is trivial.

Enumerate the nonidentity elements of Q and choose a detecting corona representation for each; pass to coordinate unitary lifts, so that each detector becomes a sequential almost representation with its marked element separated from 1. Diagonalize at the coordinate level: at stage k, select in each of the first k detectors a coordinate at which multiplicativity holds within $1/k$ on the first k group elements while the marked separation persists, and take the block sum of the k selected matrices. Operator norms of block sums are maxima, so the resulting single sequence is an almost representation detecting every nonidentity element; its class in the unitary-sequence quotient, composed with the polar-correction isomorphism of Lemma 4.2, is a faithful corona representation of Q. This proves (2). Composing it with $G_1 \twoheadrightarrow Q$ gives a single corona representation of G_1 with kernel exactly R, proving (1). For the final equivalence: if G_1 is MF, an injective corona representation forces $R = \{1\}$; if $R = \{1\}$, then (2) says $G_1 = G_1/R$ is MF. □

Corollary 10.4 (exact radical from a candidate quotient)**.** *Let $N \trianglelefteq G_1$ satisfy $N \leq \mathrm{Rad}_{\mathrm{MF}}(G_1)$. Every corona representation of G_1 factors uniquely through G_1/N. If in addition G_1/N is MF, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G_1) = N.$$

[formalized in Lean]

Together with Proposition 10.3, this says that proving a concrete defect subgroup invisible and its quotient MF computes both the entire MF radical and the universal MF quotient.

Proof. The containment $N \leq \ker \Theta$ gives the unique quotient factorization for every corona representation Θ. For the reverse containment in the displayed equality, compose the quotient map $G_1 \to G_1/N$ with an injective corona representation of G_1/N: its kernel is exactly N, while the MF radical lies in the kernel of every corona representation. □

Corollary 10.5. *Let A be a unital C^*-algebra admitting an injective, possibly nonunital, $*$-homomorphism into a norm matrix corona. Then the group E admits no injective homomorphism into $\mathcal{U}(A)$.* [formalized in Lean]

Proof. Let $j\colon A \to Q$ be the possibly nonunital inclusion into the ambient corona and let $\phi\colon E \to \mathcal{U}(A)$. Complement correction gives

$$\widehat{j}(u) = j(u) + 1 - j(1_A) \in \mathcal{U}(Q).$$

This is an injective homomorphism on $\mathcal{U}(A)$, so $\widehat{j} \circ \phi$ is a corona representation of E. Theorem A(2) sends w to the identity there; injectivity of $\widehat{j}$ gives $\phi(w) = 1$. Proposition 6.3 has $w \neq 1$, so ϕ cannot be injective. □

10.2 A finite universal Horn obstruction Theorem A can be read as one first-order fact about all MF groups at once. Write R for the forty-one relators of (9), regarded as words in eight variables, and w for the marked word in those variables.

Proposition 10.6 (finite quasi-identity)**.** *Every MF group G satisfies the finite universal Horn sentence*

$$\forall x_1, \ldots, x_8\colon \quad \bigwedge_{r \in R} r(x_1, \ldots, x_8) = 1 \implies w(x_1, \ldots, x_8) = 1,$$

while the canonical generating tuple of E satisfies every relator in R and has $w \neq 1$. [formalized in Lean]

Proof. A tuple satisfying the relators induces a homomorphism $E \to G$ sending the marked word of E to $w(x_1, \ldots, x_8)$. If G is MF, then Lemma 10.2 and Theorem A make that image trivial. The canonical tuple of E satisfies R by definition, and its marked word is nontrivial by Theorem A(1). □

Corollary 10.7 (a clopen non-MF cylinder)**.** *In the space of eight-generator marked groups — quotients of the free group F_8, topologized so that a basic clopen set fixes the truth of finitely many equations among words in the generators — the marked groups satisfying all relators of R and $w \neq 1$ form a nonempty clopen set consisting entirely of non-MF groups.* [formalized in Lean]

Proof. Both conditions constrain finitely many values of the marked multiplication, so the set is clopen; it contains the marked group E by Theorem A(1), and Proposition 10.6 excludes every MF point. ☐

Thus the obstruction of Theorem A is finite, local, and stable under perturbation in marked-group space.

10.3 Permanence and its failure

Lemma 10.8 (subgroups)**.** *Every subgroup of an MF group is MF.* [formalized in Lean]

Proof. Restrict an injective corona representation to the subgroup: injectivity and the homomorphism law are preserved. ☐

Lemma 10.9 (residually finite groups)**.** *Every countable residually finite group is MF.* [formalized in Lean]

Proof. Residual finiteness makes the group LEF: for each finite test set F there are a finite quotient injective on F and, composing with its left regular permutation action, an *exact* finite local model — every product of elements of F is represented with zero multiplicative defect. Distinct elements of F map to distinct permutations; at a point where the two permutations disagree, the difference of the two permutation matrices has a column of norm at least 1, so the model separates distinct elements of F by operator norm at least 1. These exact permutation models are operator-norm local models with separation constant 1, so the group is MF by the local-to-corona direction of Proposition 1.1. ☐

Lemma 10.10 (locally finite groups)**.** *Every countable locally finite group is MF.* [formalized in Lean]

Proof. The proof is local: no exhaustion, direct limit, or diagonal argument is needed. Given a finite test set F, local finiteness supplies one finite subgroup K containing it. The left regular permutation action of K, applied to a retraction of the group onto K that is only required to restrict to the identity map on the test set, is an exact finite local model on F, separating distinct elements by operator norm at least 1 exactly as in Lemma 10.9. Thus the group is LEF, and the same local-to-corona direction of Proposition 1.1 makes it MF. ☐

Corollary 10.11 (quotients)**.** *MF groups are not closed under quotients: the free group F_8 is MF, and the literal group E is its quotient but is not MF.* [formalized in Lean]

Proof. Map the eight free generators onto the eight displayed generators of E. The resulting homomorphism is surjective because those generators define the literal presentation. The group F_8 is residually finite, hence MF by Lemma 10.9. Theorem A shows that E is not MF. □

10.4 A sofic non-MF group The Clifford witness has substantially more structure than was needed to prove $w \neq 1$.

Proof of Theorem E. Write $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$, so $V = T \rtimes \mathbb{Z}$, and let $B \leq V$ be the level-zero copy of $\bar{\Gamma}$. The image of $\bar{\alpha}$ has finite index (at most eight, by the parity classes of the translation lattice). Thus every telescope level $\bar{\Gamma}_n \leq T$ is commensurable with B. The stable letter conjugates B onto $\bar{\alpha}(\bar{\Gamma})$, so all of V commensurates B. Orbit–stabilizer now shows that every $\bar{\Gamma}_n$-orbit in $X = V/B$ is finite.

Consider a finite subset of $\mathsf{ClLamp}(X) \rtimes T$. Its T-coordinates lie in one level $\bar{\Gamma}_n$. Each of its lamp coordinates has finite $\bar{\Gamma}_n$-orbit: this holds for the central sign and for each site lamp, hence for every Clifford word. The finitely many resulting orbits generate a finite $\bar{\Gamma}_n$-invariant subgroup $L \leq \mathsf{ClLamp}(X)$, because $\mathsf{ClLamp}(X)$ is locally finite. Our original finite set therefore lies in $L \rtimes \bar{\Gamma}_n$, which is sofic (finite-by-sofic). Soficity is local over finite subsets, so $\mathsf{ClLamp}(X) \rtimes T$ is sofic.

Reassociating the semidirect products gives

$$W = \mathsf{ClLamp}(X) \rtimes (T \rtimes \mathbb{Z}) \cong (\mathsf{ClLamp}(X) \rtimes T) \rtimes \mathbb{Z}.$$

The final group is sofic because a split extension of a sofic group by $\mathbb{Z}$ is sofic: truncate the integer levels to $\mathbb{Z}/L$, let the shift act cyclically, and induce the kernel's sofic approximations over the level set; the cocycle identity that makes the finite-quotient case exact now fails only on the wrap-around boundary, a fraction of at most $2|k|/L$ of the levels for a window shifted by k, which vanishes as L grows. This is the special case of the Elek–Szabó amenable-extension theorem [ES06] that the argument needs, proved directly. Here $\bar{\Gamma}$ is sofic because its six generators are casts of integral matrices: the integral subgroup is residually finite through its congruence quotients, and casting is an isomorphism onto $\bar{\Gamma}$.

The map $E \to W$ of Proposition 6.3 is surjective: its image contains V, the root lamp c_o, and hence every lamp by transitivity of the V-action on X. Since E is generated by v_1, x, y, z, t, c (by the Tietze calculation above), their six images generate W. The map sends $w \in \mathrm{Rad}_{\mathrm{MF}}(E)$ to the nontrivial sign ζ; portability (Lemma 10.2) makes $\zeta \in \mathrm{Rad}_{\mathrm{MF}}(W)$. Hence W is not MF. □

Failure of extension permanence. MF groups are not closed under extensions. More precisely, W fits into an exact sequence

$$1 \longrightarrow \mathsf{ClLamp}(X) \longrightarrow W \longrightarrow V \longrightarrow 1$$

whose kernel and quotient are MF, while W is not. The concrete lamp kernel is locally finite, LEF, sofic, and MF, while the total witness is finitely generated and non-MF. [formalized in Lean]

Proof. The Clifford lamp group is locally finite, hence MF by Lemma 10.10. The group V identifies with the subgroup of $\mathrm{GL}_4(\mathbb{Q})$ generated by $\bar{\Gamma}$ and D: the telescope levels are $D^{-n}\bar{\Gamma}D^n$, and the determinant records the $\mathbb{Z}$-coordinate. Its entries are dyadic rationals, so reduction modulo odd prime powers separates nontrivial elements, and V is residually finite — the conclusion Mal'cev's theorem [Mal] gives for every finitely generated linear group, obtained here by the direct congruence argument. Thus V is MF by Lemma 10.9, whereas W is not MF by the preceding construction. □

An exact stably finite non-MF algebra. The same witness gives a concrete separable exact example: $C^*_{\mathrm{red}}(W)$ is exact, has a faithful tracial state, is stably finite, and is not MF. Indeed, the locally finite lamp kernel is amenable and hence exact, while the linear group $V \leq \mathrm{GL}_4(\mathbb{Q})$ is exact [GHW]. Exact groups are closed under extensions [KW2], so W is exact; for discrete groups this is equivalent to exactness of the reduced group algebra [KW1]. The canonical trace and stable finiteness follow as in Lemma 10.12. Finally, an MF embedding of $C^*_{\mathrm{red}}(W)$ would restrict to a faithful norm-corona model of the canonical group unitaries, contradicting that W is not MF. All parts of this paragraph except exactness are also proved by a closed Lean endpoint; the present formal library has no definition of exact C^*-algebras.
A simple sofic envelope. There is a countable simple sofic group S_{simp} with

$$\mathrm{Rad}_{\mathrm{MF}}(S_{\mathrm{simp}}) = S_{\mathrm{simp}}.$$

Equivalently, every homomorphism from S_{simp} to an MF group is trivial.

Proof. By Elek–Szabó, every countable sofic group embeds in a countable simple sofic group [ES05, Theorem 1]. Embed W in such a group. Portability sends its nontrivial radical element ζ to a nontrivial element of the MF radical of the simple group. That radical is normal, so it is the whole group. Every homomorphism to an MF group kills the MF radical. (The embedding theorem is a literature input; this paragraph is outside the formal development.) □

10.5 The reduced group algebra of E

Lemma 10.12. *Let G_1 be a countable discrete group.*

1. *The canonical tracial state τ on $C^*_{\mathrm{red}}(G_1)$ is faithful.*
2. *Every unital C^*-algebra A with a faithful tracial state is stably finite: for every nonempty finite index set I, every isometry in $M_I(A)$ is a*

unitary. In particular this holds in $M_k(A)$ for every $k \geq 1$. [formalized in Lean]

Proof. (1) Realize $C^*_{\mathrm{red}}(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g, and let ρ_g be the right translation unitaries, $(\rho_g \xi)(h) = \xi(hg)$; each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C^*_{\mathrm{red}}(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e\rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}} x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k}\sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k}\sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i-th diagonal entry of x^*x is $\sum_j x^*_{ji}x_{ji}$, so $\tau_k(x^*x) = 0$ forces $\tau(x^*_{ji}x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j. If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. □

Proof of Theorem D. $C^*_{\mathrm{red}}(E)$ is unital and separable since E is countable; by Lemma 10.12 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A. □

11 The cyclic-base calibration

Proof of Theorem C. As with w, the relator $w^2_{\mathrm{BS}} = 1$ is redundant — tct^{-1} and $\gamma_0(tct^{-1})\gamma_0^{-1}$ are involutions, so centrality forces it by (5) — and is retained only to display the marked pattern. Deleting it gives a six-relator presentation of the same group. The stable relation gives $t\langle\gamma_0\rangle t^{-1} \subseteq \langle\gamma_0\rangle$, and c centralizes this cyclic base. The same commutant argument as in Theorem B therefore maps w_{BS} to the identity in every finite-dimensional representation over every field.

For nontriviality, use the affine realization of $\mathrm{BS}(1,2)$ and its coset action. The root site is fixed by the cyclic base, while the two sites represented by t and $\gamma_0 t$ are distinct. Attach Clifford generators to these sites and map c to the root lamp. All seven relators hold, and

$$w_{\mathrm{BS}} \longmapsto [c_{t\langle\gamma_0\rangle}, c_{\gamma_0 t\langle\gamma_0\rangle}] = -1.$$

Thus $w_{\mathrm{BS}} \neq 1$. Restricting the target to the subgroup generated by the three displayed images gives the asserted surjective realized quotient. This comparison isolates the precise limitation of the finite-dimensional argument: exact finite-dimensional invisibility does not itself imply that the marked word is trivial. □

Sharpness of the Kazhdan hypothesis. The realized Clifford quotient in Theorem C is amenable, hence MF, and its quotient map detects w_{BS}. In particular, $w_{\mathrm{BS}} \notin \mathrm{Rad}_{\mathrm{MF}}(E_{\mathrm{BS}})$, so property (T) cannot be omitted from the central-sign criterion. The affine group $\mathrm{BS}(1,2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ is solvable. The Clifford lamp group on its coset space is locally finite, so their semidirect product is amenable. The realized quotient is a subgroup of this amenable group and is therefore amenable, hence MF by [TWW]. Its marked word is the nontrivial Clifford sign by the calculation above. The amenable-implies-MF step is a literature input, outside the formal development; the non-LEF statement below is formalized.

This MF detector is necessarily genuinely approximate: neither the literal cyclic group nor its realized Clifford quotient is LEF, and the latter is not residually finite. Indeed, in every finite group the doubling relation makes the image of $\langle \gamma_0^2 \rangle$ equal to the image of $\langle \gamma_0 \rangle$; the lamp therefore commutes with the required "halved" element and the marked word collapses. [formalized in Lean]

12 The operator-norm boundary

The proof is genuinely operator-norm in its present form. If multiplication is controlled only in normalized Hilbert–Schmidt norm, the induced adjoint operators on the Hilbert–Schmidt spaces need not be close in operator norm. The spectral-corner, displacement, and leakage estimates of Theorem 2.1 require exactly that operator-norm control, so normalized Hilbert–Schmidt control does not supply the spectral-subspace transport used there. This is a description of where the printed estimates need their hypotheses, not a theorem that no tracial variant can exist.

This identifies a boundary of the method, not a tracial nonapproximation theorem for E. Alekseev–Thom ask whether commutants arising in sofic approximations of Kazhdan groups can be recovered from finite-dimensional coordinate subalgebras [AT, Open Problem 6.2]. Such a coordinate description is separate from the operator-norm theorem proved here.

Questions The finite-normal mechanism has a precise torsion limitation: every finite subgroup of a torsion-free group is trivial. [formalized in Lean] Consequently a torsion-free example would require a different final obstruction, even if its transport step were similar. Fournier-Facio removed the analogous torsion obstacle for nonsoficity by a small-cancellation construction [FFF]. That strategy does not transfer directly here: the final operator-norm contradiction uses a nontrivial finite normal subgroup, which a torsion-free group cannot contain.

(1) Is $E/\langle w \rangle$ MF? By Corollary 10.4, a positive answer would compute $\mathrm{Rad}_{\mathrm{MF}}(E) = \{1, w\}$ exactly. There is a concrete intermediate test. Theorem B kills $u = [tct^{-1}, v_1]$ in every finite-dimensional representation, whereas the image of u survives after quotienting the Clifford witness W by its central sign ζ: it becomes the product of the two distinct lamp basis elements at τo and $j(\bar{v}_1)\tau o$. Thus proving $W/\langle \zeta \rangle$ MF would already show $u \notin \mathrm{Rad}_{\mathrm{MF}}(E)$; any such detecting models must be genuinely non-finite-quotient in flavor.

(2) Is there a torsion-free finitely presented non-MF group? The mechanism here is torsion-critical: the obstruction lives in a finite normal subgroup.

(3) Is E sofic, or hyperlinear? The adjoint Hilbert-ultraproduct representation requires operator-norm multiplicative control, and neither property is decided by the results above.
(4) Does MF imply hyperlinearity? A faithful embedding into a norm matrix corona need not make any coordinate trace faithful, so the defining MF representation does not immediately supply a hyperlinear model. This question concerns the CDE norm-corona convention fixed above; for stronger trace-controlled MF conventions the implication is built into the definition.

A Proof of the MF equivalences

An *operator-norm local model* for a countable group E, a finite set $F \subseteq E$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$ is a map $V: E \to \mathcal{U}(r)$, for some positive dimension r, such that

$$\|V(g)V(h) - V(gh)\| \leq \varepsilon \quad (g, h \in F),$$
$$\|V(g) - V(h)\| \geq \delta \quad (g \neq h \in F).$$

The local formulation in Proposition 1.1 asks for such a model with $\delta = 1$ for every finite F and every $\varepsilon > 0$.

Proof of Proposition 1.1. Lemma 4.2 identifies the unitary-sequence quotient with the unitary group of the actual C^*-quotient. Replacing an arbitrary finite coordinate set by a standard set of the same cardinality merely conjugates that coordinate by a permutation unitary. This proves the first equivalence once the dimension convention is normalized.

Suppose first that $\rho: G \to \mathsf{U}_{\mathrm{cor}}((d_n))$ is injective, and choose unitary lifts $u_n(g)$ of $\rho(g)$. Fix a finite set $F \subseteq G$ and $\varepsilon > 0$. For each ordered pair $g \neq h$ in F, injectivity says that $(u_n(g)^{-1}u_n(h))_n$ does not converge to the identity in operator norm. Hence there are $\delta_{g,h} > 0$ and arbitrarily large coordinates at which

$$\|u_n(g) - u_n(h)\| \geq \delta_{g,h}.$$

Choose an integer $N_{g,h}$ with $8 < N_{g,h}\, \delta_{g,h}^2$. An explicit tensor-power estimate then applies: whenever two unitaries are $\delta_{g,h}$-separated, some tensor power of exponent $p \leq N_{g,h}$ separates them by more than 1 — amplifying a spectral gap of the relative unitary until it crosses the unit circle takes at most $N_{g,h}$ steps once $8 < N_{g,h}\delta_{g,h}^2$. The quotient homomorphism law makes all multiplicative defects for pairs in F arbitrarily small at every sufficiently large coordinate. Choose, for each $g \neq h$, a coordinate with the displayed separation and with every required defect at most $\varepsilon/N_{g,h}$; then the pth tensor power separates the designated pair by more than 1, while the tensor-product telescoping identity bounds each new defect by $p\,\varepsilon/N_{g,h} \leq \varepsilon$. Finally take the block sum of these finitely many pair-specific models. Operator norms of block diagonal matrices are maxima, so every pair in F is separated by at least 1 and every required product has defect at most ε. This gives the normalized local model.

Conversely, local models with separation constant 1 for all finite sets and tolerances make the group *weakly* MF: choosing models along an exhaustion with errors tending to zero, each fixed pair g, h is eventually represented multiplicatively and each fixed pair $g \neq h$ eventually separated by 1, so the classes define an injective homomorphism into the unitary-sequence quotient with no condition on the dimension sequence. Weak MF in turn makes the norm-MF radical of Definition 10.1 trivial, and the block-sum diagonalization in the proof of Proposition 10.3(2) then assembles the detecting sequences into a single faithful corona representation. This is the actual dependency route: the weak-to-genuine upgrade is the universal-quotient construction, not a separate exhaustion.

It remains only to normalize dimensions. Given positive dimensions d_n, replace the nth coordinate model by the cumulative block sum on $X_0 \oplus \cdots \oplus X_n$, where X_k is the kth original model space: embed the nth original model in the newest block and put identity blocks in all earlier positions. The dimensions $D_n = d_0 + \cdots + d_n$ increase strictly. Products, multiplicative defects, and pairwise operator-norm distances are unchanged, because block-diagonal operator norms are maxima and every earlier block is the same identity for all group elements; in particular no approximation error from earlier coordinates is carried forward. Thus the construction preserves and reflects the quotient relation and injectivity. Forgetting strict increase proves the converse. □

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